Answer key - Solving Quadratic Equations

GCE Ordinary Level - Mathematics - Corrige

Detailed method for every exercise. Follow the substitutions line by line.
Corrige detaille de chaque exercice. Suis les substitutions ligne par ligne.

1. Solve (x + 5)(x - 2) = 0.
Résous (x + 5)(x - 2) = 0.

- x = -5 or x = 2 / x = -5 ou x = 2 *
- x = 5 or x = -2 / x = 5 ou x = -2
- x = 5 or x = 2 / x = 5 ou x = 2
- x = -5 or x = -2 / x = -5 ou x = -2

Answer: x = -5 or x = 2
x + 5 = 0 -> x = -5; x - 2 = 0 -> x = 2. The roots are -5 and 2.
x + 5 = 0 -> x = -5; x - 2 = 0 -> x = 2. Les racines sont -5 et 2.

2. Solve x^2 + 5x + 6 = 0.
Résous x^2 + 5x + 6 = 0.

- x = -2 or x = -3 / x = -2 ou x = -3 *
- x = 2 or x = 3 / x = 2 ou x = 3
- x = -6 or x = -1 / x = -6 ou x = -1
- x = -1 or x = 5 / x = -1 ou x = 5

Answer: x = -2 or x = -3
x^2 + 5x + 6 = (x + 2)(x + 3) = 0 -> x = -2 or -3.
x^2 + 5x + 6 = (x + 2)(x + 3) = 0 -> x = -2 ou -3.

3. Solve x^2 - 25 = 0.
Résous x^2 - 25 = 0.

- x = 5 or x = -5 / x = 5 ou x = -5 *
- x = 25 or x = -25 / x = 25 ou x = -25
- x = 0 or x = 5 / x = 0 ou x = 5
- x = 5 only / x = 5 seulement

Answer: x = 5 or x = -5
x^2 - 25 = (x - 5)(x + 5) = 0 -> x = 5 or x = -5.
x^2 - 25 = (x - 5)(x + 5) = 0 -> x = 5 ou x = -5.

4. Solve x^2 + 3x - 4 = 0 using factorisation. The larger root is a. Find a.
Résous x^2 + 3x - 4 = 0 par factorisation. La plus grande racine est a. Trouve a.

Answer: 1
x^2 + 3x - 4 = (x + 4)(x - 1) = 0 -> roots -4 and 1. Larger root is 1.
x^2 + 3x - 4 = (x + 4)(x - 1) = 0 -> racines -4 et 1. La plus grande est 1.

5. Solve (x + 4)(x - 1) = 0.
Résous (x + 4)(x - 1) = 0.

- x = -4 or x = 1 / x = -4 ou x = 1 *
- x = 4 or x = -1 / x = 4 ou x = -1
- x = 4 or x = 1 / x = 4 ou x = 1
- x = -4 or x = -1 / x = -4 ou x = -1

Answer: x = -4 or x = 1

x + 4 = 0 -> -4; x - 1 = 0 -> 1.

6. Solve x^2 - 49 = 0.

Résous x^2 - 49 = 0.

- x = 0
- x = 7 or x = -7 / x = 7 ou x = -7 *
- x = 49 or x = -49 / x = 49 ou x = -49
- x = 7 only / x = 7 seulement

Answer: x = 7 or x = -7

(x - 7)(x + 7)=0 -> ±7.

7. Solve x^2 + x - 6 = 0.

Résous x^2 + x - 6 = 0.

- x = -3 or x = 2 / x = -3 ou x = 2 *
- x = 3 or x = -2 / x = 3 ou x = -2
- x = -6 or x = 1 / x = -6 ou x = 1
- x = 6 or x = -1 / x = 6 ou x = -1

Answer: x = -3 or x = 2

Factorise to (x + 3)(x - 2)=0 -> x = -3 or 2.

Factorise en (x + 3)(x - 2)=0 -> x = -3 ou 2.

8. Solve x^2 - x - 6 = 0. The positive root is a. Find a.

Résous x^2 - x - 6 = 0. La racine positive est a. Trouve a.

Answer: 3

Factorise (x - 3)(x + 2)=0 -> roots 3 and -2. Positive root is 3.

Factorise (x - 3)(x + 2)=0 -> racines 3 et -2. La racine positive est 3.

9. Solve x^2 - 9 = 0. The larger root is a. Find a.

Résous x^2 - 9 = 0. La racine la plus grande est a. Trouve a.

Answer: 3

(x - 3)(x + 3)=0 -> roots 3 and -3. Larger is 3.

(x - 3)(x + 3)=0 -> racines 3 et -3. La plus grande est 3.

Mega Mighty Sixers - contenu original, diffusion libre aux membres de l association